

جـامعة بيروت العربية
BEIRUT ARAB UNIVERSITY

Faculty of Science

Department of Mathematics and Computer Science

Computer Security (CMPS455)

Cipher Forensics and Attack Recommendation Engine

Student Name:	**Student ID:**
Nour Ajami	202302485

Supervisor:

Dr. Mohamed Al Saleh	Supervisor
T.A. Lynn El Arwadi	Lab Instructor

April 30, 2026

Table of Contents

MAIN PROBLEM DESCRIPTION 4
MOTIVATION 4
PROJECT OBJECTIVE 5
CONTRIBUTION OF THE PROJECT 5
CRYPTOANALYSIS AND CIPHER DETECTION 6
RELATED ATTACK METHODS 6
SUPPORT CIPHER TYPES 7
PATTERN ANALYSIS 7
FREQUENCY ANALYSIS 7
PROPOSED ALGORITHM 8
WHY THIS ALGORITHM? 8
ALGORITHM STEPS 8
SYSTEM WORKFLOW 8
TOOLS AND LIBRARIES 9
TESTING PROCEDURE 9
EVALUATION APPROACH 9
SYSTEM OUTPUT 10
CIPHER DETECTION RESULTS 10
ANALYSIS OF RESULTS 10

Abstract

Cryptography is used to secure and protect sensitive information and messages. Most encryption can be cryptanalyzed to figure out the specific algorithm or method that was used to encrypt the messages in the first place. This project is based on a Cipher Forensic and Attack Recommendation Engine which identifies and recommends the most likely family of encryption of a message. The system covers three of the main types of classical ciphers (Caesar, monoalphabetic substitution and transposition ciphers) and gives both a pattern and frequency analysis of the given ciphertext, reasoning as to why that cipher was chosen for encryption and advice on the best method to use to decrypt the message.

This project is a demonstration of a complete cryptoanalysis system, which does detection, reasoning, and attack planning all in one. I implemented this project using Python language because python is the easiest language to process text strings.

Chapter 1. Introduction

Main Problem Description

Encrypted communication is a powerful tool in the field of computer security. Once an encrypted message has been intercepted by an attacker it becomes an important piece of information. By identifying the encryption used to scramble the message, information security professionals can either use it to their benefit or waste it.

Ciphers and encryption techniques all have distinct patterns in their encrypted forms. Some encrypted text patterns include letter frequency, patterns of letters rearranged or shifted, or simply substituted for other letters. Understanding these patterns and where to find them is crucial in the world of cryptoanalysis.

This project aims to analyze given ciphertext, guess the most likely cipher to have been used, and then simulate an attack to suggest a course of action given the family of the ciphertext's encryption.

Motivation

The motivation for this project is to investigate how cryptanalysis systems operate in practice, specifically looking at message analysis techniques which utilize statistical properties and pattern recognition, and do not simply rely on the ability to encrypt and decrypt.
To enhance the understanding of the project, I implemented the basic concepts of computer security. The program analyses a given text, makes logical conclusions and in the end, it even gives suggestions for potential attacks.
The system is intended to simulate the first steps of real cryptanalysis.

Project Objective

This project attempts to create a system able to analyze ciphertext and determine which family of encryption it belongs to.

The project objectives are:

- Analyze ciphertext using frequency analysis.
- Detect the most likely cipher type.
- Explain why the system selected that cipher.
- Recommend the most suitable attack strategy.
- Demonstrate basic cryptoanalysis concepts.

Contribution of the Project

This project combines several important concepts of computer security into one application. The system does not simply classify ciphertext, it also explains the reasoning behind its decision and suggests possible attack methods.
The project demonstrates:

- Pattern analysis
- Frequency analysis
- Cipher detection
- Security reasoning
- Attack recommendation

This creates a complete workflow like practical cryptanalysis systems.

Chapter 2. Related Work

Cryptoanalysis and Cipher Detection

Cryptanalysis is the attempt to guarantee security provisions of encrypted messages. Many of the classical ciphers can be identified by the study of their statistical and structural properties.

Frequency analysis is a common method for cryptanalysis, based on the observed fact that some letters appear more frequently in English than others. Frequency analysis attacks compare the letter frequency of the ciphertext against know letter for common languages.

The classical ciphers like the Caesar cipher, classical substitution ciphers and transposition ciphers have very different characteristics. So, it is possible to estimate the encryption family used.

Related Attack Methods

Brute Force Attack

A brute force attack tries every possible key until the message is decrypted. This attack is commonly used against Caesar ciphers because they only have 25 possible shifts.

Frequency Analysis

The Frequency analysis compares the letter frequencies of plain text and the ciphertext. This test compares the letter frequencies of typical English letters E, T, A, O. This technique is commonly used against substitution ciphers.

Structural Analysis

Structural analysis studies the arrangement and ordering of characters. It is commonly used to analyze transposition ciphers where the letters of the plaintext are rearranged rather than replaced with different letters.

Chapter 3. System Description

Support Cipher Types

The system uses three classical encryptions:

- Caesar Cipher
- Monoalphabetic Substitution Cipher
- Transposition Cipher

Pattern Analysis

The system analyzes ciphertext patterns by examining:

- Number of unique letters.
- Distribution of letters.
- Repetition frequency.

Frequency Analysis

This program computes the letter frequencies in the given ciphertext and uses them during the cryptanalysis.

For example:

- Small variation in letters suggest Caesar cipher.
- Moderate variation suggests substitution cipher.
- High variation suggests transposition cipher.

Chapter 4. Methodology

Proposed Algorithm

In this report, I proposed rule-based cryptanalysis, and use simple rules based on letter frequency and patterns to determine the type of the cipher.

Why This Algorithm?

I used a simple cipher to enable demonstration of important cryptographic concepts. The system is simple enough and sufficiently well understood that it focuses on learning the reasoning behind about how to attack the cipher and suggesting attacks rather than decrypting messages.

This solution can be implemented in many languages and also shown how it can be done with Python, and it covers some of the main concepts of computer security.

Algorithm Steps

- Receive ciphertext input from user.
- Analyze letter frequency.
- Count unique letters.
- Compare patterns using predefined rules.
- Estimate cipher type.
- Explain reasoning.
- Recommend attack strategy.

System Workflow

Ciphertext is included into the system and then analyzed using frequency analysis and pattern analysis within the system. The system then analyzes the results to determine the most likely encryption family and provides an explanation for its answer. The system also examines the characteristics of the ciphertext and provides the user with a recommended strategy for attack.

Chapter 5. Experimental Setup

Tools and Libraries

The project was implemented using python.

The following tools were used:

- Visual Studio Code
- Python 3
- Terminal

Testing Procedure

To test the function of the system, several well known and unknown ciphertexts including few test files were entered. As ciphertexts with different characteristics were entered, the system classified the encryption family by observing the length of the ciphertext.

Evaluation Approach

The system was evaluated using:

- Correctness of cipher detection
- Quality of explanation
- Suitability of attack recommendation

Chapter 6. Results and Analysis

System Output

Example Input:

```
===Cipher Forensics and Attack Recommendation Engine===
Enter ciphertext: KHOOR ZRUOG
```

Example Output:

```
===Pattern Analysis===
Letter Frequency: {'K': 1, 'H': 1, 'O': 3, 'R': 2, 'Z': 1, 'U': 1, 'G': 1}
Number of Unique Letters: 7

===Estimated Encryption Family===
Detected Cipher: Caesar Cipher

===Explanation===
The ciphertext contains a small number of unique letters. This means that it is a simple shift pattern, which is Caesar cipher.

===Recommended Attack Strateg===
Use brute-force attack by trying all 25 possible shifts.
```

Cipher Detection Results

Different encryption families were identified through frequency and patterns in the data and explanations were provided for the system's decision.

Analysis of Results

In addition to detailing how certain simple statistical analysis can reveal information about encrypted text, the system only uses two simplified rules for encryption.

Chapter 7. Discussion

This project tries to integrate pattern analysis, reasoning, and attack recommendations into a single workflow. It shows how analyzing ciphertext can be used to identify the encryption methods that can be attacked before even attempting any decryption.

Keeping rules simple was a challenge, but sometimes the results just weren't reasonable. Also, between simplicity and accuracy, simplicity won.

The project supports only classical ciphers but provide good understanding of practical cryptanalysis concepts.

Chapter 8. Conclusion

This project implemented a Cipher Forensics and Attack Recommendation Engine that analyzes given ciphertext and estimates the most likely encryption family and provides guidance on how to launch an attack against it.

The system successfully demonstrated:

- Frequency analysis
- Pattern Recognition
- Cipher detection
- Attack recommendation

The project demonstrates the use of reasoning and analysis in computer security and cryptanalysis.

Chapter 9. Future Work

Several improvements can be added in future versions:

- Support additional cipher types.
- Add automatic decryption.
- Use machine learning for better detection.
- Develop graphical user interface.
- Improve accuracy using advanced statistical methods.

Chapter 10. References

- **https://csrc.nist.gov/**

- **https://www.ibm.com/think/topics/cryptography**

- **https://www.geeksforgeeks.org/ethical-hacking/caesar-cipher-in-cryptography/**

- **https://www.britannica.com/topic/cryptology**

Chapter 11. Appendix

Full Code:

```
def analyze_frequency(text):
    frequency = {}

    for char in text:
        if char.isalpha():
            char = char.upper()
            frequency[char] = frequency.get(char, 0) + 1

    return frequency


def detect_cipher(text):
    freq = analyze_frequency(text)
    unique_letters = len(freq)

    if unique_letters < 10:
        return (
            "Caesar Cipher",
            "The ciphertext contains a small number of unique letters. "
            "This means that it is a simple shift pattern, which is Caesar cipher."
        )

    elif unique_letters <= 20:
        return (
            "Monoalphabetic Substitution Cipher",
            "The ciphertext shows moderate variation in letters. "
            "This means that the letters are replaced but patterns still exist, which is substitution ciphers."
        )

    else:
        return (
            "Transposition Cipher",
            "The ciphertext contains many different letters. "
            "This means that the letters are rearranged rather than replaced, which is transposition ciphers."
        )
```

```python
def recommend_attack(cipher):
    if cipher == "Caesar Cipher":
        return "Use brute-force attack by trying all 25 possible shifts."

    elif cipher == "Monoalphabetic Substitution Cipher":
        return "Use frequency analysis by comparing common letters such as E, T, A, and O."

    else:
        return "Use structural analysis by rearranging positions and testing possible patterns."


def main():
    print("===Cipher Forensics and Attack Recommendation Engine===")

    text = input("Enter ciphertext: ")

    freq = analyze_frequency(text)
    cipher, explanation = detect_cipher(text)
    attack = recommend_attack(cipher)

    print("\n===Pattern Analysis===")
    print("Letter Frequency:", freq)
    print("Number of Unique Letters:", len(freq))

    print("\n===Estimated Encryption Family===")
    print("Detected Cipher:", cipher)

    print("\n===Explanation===")
    print(explanation)

    print("\n===Recommended Attack Strateg===")
    print(attack)


main()
```

Output:

```
===Cipher Forensics and Attack Recommendation Engine===
Enter ciphertext: KHOOR ZRUOG

===Pattern Analysis===
Letter Frequency: {'K': 1, 'H': 1, 'O': 3, 'R': 2, 'Z': 1, 'U': 1, 'G': 1}
Number of Unique Letters: 7

===Estimated Encryption Family===
Detected Cipher: Caesar Cipher

===Explanation===
The ciphertext contains a small number of unique letters. This means that it is a simple shift pattern, which is Caesar cipher.

===Recommended Attack Strateg===
Use brute-force attack by trying all 25 possible shifts.
```